ALGEBRA 1 · SYSTEMS OF EQUATIONS AND INEQUALITIES

Systems Explorer

Explore & Reflect **TEACHER KEY**

Name _____ Date _____ Period _____

Open **Systems Explorer** on Mr. K's Classroom: Algebra 1 → Systems of Equations and Inequalities → Interactive Math. Answer Part 1 **before** you use the tool. Use the tool for Parts 2 and 3, then do Part 4 on your own.

1 Predict before you touch the tool

1. Two lines have the same slope but different y-intercepts. How many times will they cross?

   Never. They are parallel.

2. What does the point where two lines cross tell you?

   It is the (x, y) pair that makes both equations true.

2 Explore use the tool and record what you see

Equations

Set each system with the sliders.

Line 1	Line 2	What the graph shows	Number of solutions	Solution
$y = x + 1$	$y = -x + 5$	**the lines cross once**	**1**	**(2, 3)**
$y = 2x - 1$	$y = 2x + 3$	**parallel lines**	**0**	**none**
$y = -\frac{1}{2}x + 4$	$y = -\frac{1}{2}x + 4$	**the same line**	**infinitely many**	**every point on the line**
$y = 3x - 2$	$y = x + 2$	**the lines cross once**	**1**	**(2, 4)**

Inequalities: $y \geq x + 1$ and $y \leq -x + 5$

Switch to Inequalities mode and drag the test point.

Test point	Line 1 true?	Line 2 true?	Solution of the system?
(0, 0)	**no**	**yes**	**no**
(1, 4)	**yes**	**yes**	**yes**
(4, 0)	**no**	**yes**	**no**
(0, 3)	**yes**	**yes**	**yes**

3 Write the rule fill in each blank from what you observed

1. The solution of a system is the point where the lines **intersect** .
2. Same slope and different y-intercepts: **no** solution.
3. Same slope and same y-intercept: **infinitely many** solutions.
4. Different slopes: exactly **one** solution.
5. For a system of inequalities, the solutions are the points where the shadings **overlap** .

4 Check no tool: show your work

1. Solve $y = 2x + 1$ and $y = -x + 7$.

   (2, 5)

2. How many solutions: $y = 4x - 3$ and $y = 4x + 1$?

   None

3. How many solutions: $y = 3x + 6$ and $y = 3(x + 2)$?

   Infinitely many

4. Is (1, 2) a solution of $y > x$ and $y \leq 3$?

   Yes: $2 > 1$ and $2 \leq 3$

5 Reflect

How can you tell how many solutions a system has without graphing?

Compare slopes and y-intercepts. Different slopes: one solution. Same slope and different intercepts: none. Same slope and same intercept: infinitely many.